SUBJECT GUIDE

General Science

Physics, chemistry and biology at the level SSC, Railway and TET papers ask.

8	23	46
Modules	Topics	NCERT chapters

Asked in: Railway · SSC · State PSC · TET/TRE · Defence

Contents

1 Motion, force, work & energy
2 Light, sound, electricity & magnetism
3 Matter, elements & chemical reactions
4 Acids, bases, metals & carbon
5 Cell, life processes & human body
6 Reproduction, heredity & diversity
7 Health, microbes & nutrition
8 Environment & space

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MODULE 1

Motion, force, work & energy

- ☐ Speed, velocity, acceleration
- ☐ Newton's laws, friction, pressure
- ☐ Work, energy, power, simple machines

How it's asked: SI units and formulas; Railway papers love unit ↔ quantity questions.

READ (FREE, OFFICIAL)

NCERT Class 9 · Describing Motion Around Us · PDF ↗
NCERT Class 9 · How Forces Affect Motion · PDF ↗
NCERT Class 9 · Work, Energy, and Simple Machines · PDF ↗
NCERT Class 8 · Exploring Forces · PDF ↗
NCERT Class 8 · Pressure, Winds, Storms, and Cyclones · PDF ↗
NCERT Class 8 · Force and Pressure · PDF ↗
NCERT Class 8 · Friction · PDF ↗

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MODULE 2

Light, sound, electricity & magnetism

- ☐ Reflection, refraction, lenses, human eye
- ☐ Sound waves
- ☐ Current, Ohm's law, magnetic effects

How it's asked: Mirror/lens uses and defects of vision are frequent.

READ (FREE, OFFICIAL)

- NCERT Class 10 · Light – Reflection and Refraction · PDF ↗
- NCERT Class 10 · The Human Eye and the Colourful World · PDF ↗
- NCERT Class 10 · Electricity · PDF ↗
- NCERT Class 10 · Magnetic Effects of Electric Current · PDF ↗
- NCERT Class 9 · Sound Waves: Characteristics and Applications · PDF ↗
- NCERT Class 8 · Electricity: Magnetic and Heating Effects · PDF ↗
- NCERT Class 8 · Light: Mirrors and Lenses · PDF ↗

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MODULE 3

Matter, elements & chemical reactions

- ☐ States & nature of matter, mixtures
- ☐ Atoms, molecules, structure of the atom
- ☐ Chemical reactions & equations

How it's asked: Element symbols, atomic numbers and types of reactions.

READ (FREE, OFFICIAL)

- NCERT Class 9 · Exploring Mixtures and their Separation · PDF ↗
- NCERT Class 9 · Journey Inside the Atom · PDF ↗
- NCERT Class 9 · Atomic Foundations of Matter · PDF ↗
- NCERT Class 8 · Particulate Nature of Matter · PDF ↗
- NCERT Class 8 · Nature of Matter: Elements, Compounds, and Mixtures · PDF ↗
- NCERT Class 8 · The Amazing World of Solutes, Solvents, and Solutions · PDF ↗
- NCERT Class 10 · Chemical Reactions and Equations · PDF ↗

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MODULE 4

Acids, bases, metals & carbon

- ☐ pH, indicators, salts
- ☐ Metals, non-metals, corrosion
- ☐ Carbon compounds

How it's asked: Common names ↔ chemical formulas (baking soda, plaster of Paris).

READ (FREE, OFFICIAL)

NCERT Class 10 · Acids, Bases and Salts · PDF ↗

NCERT Class 10 · Metals and Non-metals · PDF ↗

NCERT Class 10 · Carbon and its Compounds · PDF ↗

NCERT Class 7 · Exploring Substances: Acidic, Basic, and Neutral · PDF ↗

NCERT Class 7 · The World of Metals and Non-metals · PDF ↗

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MODULE 5

Cell, life processes & human body

- ☐ Cell & tissues
- ☐ Nutrition, respiration, transport, excretion
- ☐ Control & coordination, hormones

How it's asked: Organ ↔ function, gland ↔ hormone.

READ (FREE, OFFICIAL)

NCERT Class 9 · Cell: The Building Block of Life · PDF ↗

NCERT Class 9 · Tissues in Action · PDF ↗

NCERT Class 10 · Life Processes · PDF ↗

NCERT Class 10 · Control and Coordination · PDF ↗

NCERT Class 7 · Life Processes in Animals · PDF ↗

NCERT Class 7 · Life Processes in Plants · PDF ↗

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MODULE 6

Reproduction, heredity & diversity

- ☐ Reproduction in plants & animals
- ☐ Heredity, Mendel
- ☐ Classification of living things

How it's asked: Scientist ↔ discovery and classification examples.

READ (FREE, OFFICIAL)

- NCERT Class 10 · How do Organisms Reproduce? · PDF ↗
- NCERT Class 10 · Heredity · PDF ↗
- NCERT Class 9 · Reproduction: How Life Continues · PDF ↗
- NCERT Class 9 · Patterns in Life: Diversity and Classification · PDF ↗
- NCERT Class 6 · Diversity in the Living World · PDF ↗

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MODULE 7

Health, microbes & nutrition

- ☐ Diseases & their causes
- ☐ Vitamins & deficiency diseases
- ☐ Microorganisms

How it's asked: Disease ↔ pathogen ↔ vaccine and vitamin ↔ deficiency disease.

READ (FREE, OFFICIAL)

- NCERT Class 8 · The Invisible Living World: Beyond Our Naked Eye · PDF ↗
- NCERT Class 8 · Health: The Ultimate Treasure · PDF ↗
- NCERT Class 8 · Microorganisms: Friend and Foe · PDF ↗
- NCERT Class 6 · Mindful Eating: A Path to a Healthy Body · PDF ↗

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MODULE 8

Environment & space

- ☐ Ecosystems, food chains, pollution
- ☐ Earth, moon, solar system, ISRO missions

How it's asked: Recent ISRO missions come from current affairs — see the capsule.

READ (FREE, OFFICIAL)

NCERT Class 10 · Our Environment · PDF ↗

NCERT Class 9 · Earth as a System: Energy, Matter, and Life · PDF ↗

NCERT Class 8 · Keeping Time with the Skies · PDF ↗

NCERT Class 8 · How Nature Works in Harmony · PDF ↗

NCERT Class 8 · Our Home: Earth, a Unique Life Sustaining Planet · PDF ↗

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Practise

Railway & SSC mocks →

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